Department of CSE(AIML), PES University
UE26CS151A -Python for Computational Problem Solving
Laboratory Week 5

Note: Students are required to write the programs along with atleast two outputs for each experiment in the laboratory observation record. The observation record must be signed by the faculty in charge. For every laboratory session, the observation component carries **marks**.

Problem Statements

Q1. Search an Element in a Tuple

Write a program to check whether an element is present in a tuple.

Q2. Remove an Element from a Set

Write a program to remove an element from a set.

Q3. Union and Sum of Unique Elements in Sets

Write a program to create two sets of integers. Find their union and calculate the sum of all unique elements in the resulting set.

Q4. Tuple and Set Operations

Write a Python program to create a tuple containing the names of six students, where some names may be repeated. Convert the tuple into a set to remove duplicate names. Display the original tuple, the resulting set, and the number of unique students.

Q5. Student Attendance Management Using Sets

Write a program to create a set containing the roll numbers of students present in a class. Remove the roll numbers that are less than a given value and display the updated set.

Q6. Common Participants in Two Events

Write a program to create two sets representing students participating in two different events. Display the names of students participating in both the events.

Q7. Student Club Membership Analysis

Write a program to create two sets representing students enrolled in two different clubs. Display the students who are common to both clubs and those who belong to only one club.

Q8. Comprehensive Set Operations

Write a program to create two sets of integers and display their union, intersection, difference, and symmetric difference.

Solutions

1. Search an Element in a Tuple

```
t = (5, 10, 15, 20)

num = int(input("Enter a number: "))

if num in t:
    print("Element found")
else:
    print("Element not found")
```

Output
Enter a number: 15
Element found

2. Remove an Element from a Set

```
s = {10, 20, 30, 40}

s.remove(20)

print(s)
```

Output
{40, 10, 30}

3. Union and Sum of Unique Elements in Sets

```
set1 = {10, 20, 30, 40}
set2 = {30, 40, 50, 60}

result = set1 | set2
```

```
print("Union:", result)

total = 0
for i in result:
    total += i

print("Sum of unique elements:", total)
```

Output

```
Union: {10, 20, 30, 40, 50, 60}
Sum of unique elements: 210
```

4. Tuple and Set Operations

```
students = ("Anil", "Bhavya", "Anil", "Charan", "Divya", "Bhavya")

unique_students = set(students)

print("Original Tuple:", students)
print("Unique Students:", unique_students)
print("Number of Unique Students:", len(unique_students))
```

Output

```
Original Tuple: ('Anil', 'Bhavya', 'Anil', 'Charan', 'Divya', 'Bhavya')
Unique Students: {'Anil', 'Bhavya', 'Charan', 'Divya'}
Number of Unique Students: 4
```

5. Student Attendance Management Using Sets

```
attendance = {101, 102, 103, 104, 105, 106, 107}

limit = int(input("Enter the minimum roll number: "))

updated_set = set()

for roll in attendance:
    if roll >= limit:
```

```
    updated_set.add(roll)

print("Updated Attendance Set:", updated_set)
```

Output

```
Enter the minimum roll number: 105
Updated Attendance Set: {105, 106, 107}
```

6. Common Participants in Two Events

```
event1 = {"Amit", "Riya", "Neha", "John", "Kiran"}
event2 = {"Neha", "John", "Sara", "Kiran", "Anu"}

common_students = event1.intersection(event2)

print("Students participating in both events:")

for student in common_students:
    print(student)
```

Output

```
Students participating in both events:
John
Neha
Kiran
```

7. Student Club Membership Analysis

```
club1 = {"Amit", "Riya", "John", "Neha"}
club2 = {"John", "Neha", "Sara", "Kiran"}

common = club1.intersection(club2)
only_one = club1.symmetric_difference(club2)

print("Students in both clubs:")
for student in common:
    print(student)

print("\nStudents in only one club:")
```

```
for student in only_one:
    print(student)
```

Output

```
Students in both clubs:
John
Neha

Students in only one club:
Amit
Riya
Sara
Kiran
```

8. Comprehensive Set Operations

```
set1 = {10, 20, 30, 40, 50}
set2 = {30, 40, 50, 60, 70}

print("Union:", set1.union(set2))
print("Intersection:", set1.intersection(set2))
print("Difference (Set1 - Set2):", set1.difference(set2))
print("Symmetric Difference:", set1.symmetric_difference(set2))
```

Output

```
Union: {10, 20, 30, 40, 50, 60, 70}
Intersection: {30, 40, 50}
Difference (Set1 - Set2): {10, 20}
Symmetric Difference: {10, 20, 60, 70}
```